Exploratory Data Analysis (EDA)

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1 Introduction

Animal movement datasets collected from satellite telemetry provide valuable records of where and when individuals are observed, but they also present practical challenges for analysis. Observations are often irregular in time, sampling density can vary substantially across individuals, and location estimates can contain heterogeneous measurement error depending on transmission conditions and quality classes. These characteristics make exploratory data analysis (EDA) an essential first step to understand the structure and limitations of the data before modeling and making inferences.

2 Data description& Overview

The cleaned whale shark dataset contains 12173 spatio-temporal location observations from 60 tagged whale sharks collected between 2011 and 2021. It contains 6 variables. id identifies the individual whale shark, date stores the observation timestamp (YYYY-MM-DD HH:MM:SS), and lat/lon give the estimated latitude and longitude in decimal degrees. There are two additional fields indicating data quality and acquisition: lc is the Argos location class, which measures error type. It reflects the expected reliability of the estimated position; numeric classes (e.g., 3, 2, 1, 0) generally correspond to decreasing accuracy, auxiliary classes such as A and B typically do not provide a quantified accuracy estimate, and Z is commonly treated as an invalid or highly unreliable location category. sat indicates which Argos satellite received the transmission.

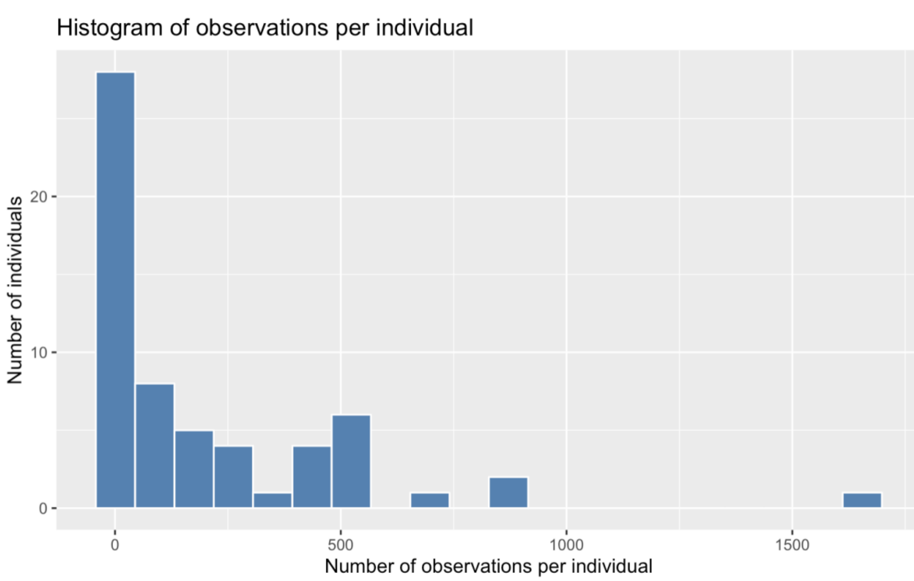

Figure 1: Sampling imbalance across tagged whale sharks.

Figure 1 shows that sampling imbalance across tagged whale sharks. The distribution of the number of observations per individual is strongly right-skewed: most individuals have smaller number of observations, while a small number of individuals have higher tracking records. The number of observations per individual ranges from 1 to 1655.

3 Distribution for Error type

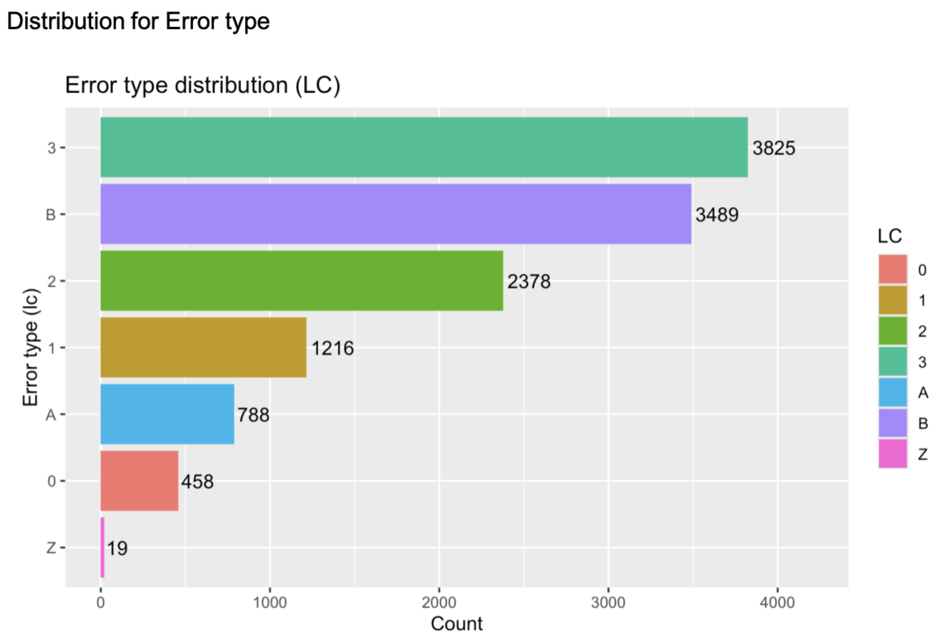

Figure 2: Distribution of Argos location classes (LC).

The barplot summarizes the distribution of Argos location classes (LC) in the dataset. In our data, the most common classes are LC 3 and LC B. Lower-quality or unquantified classes appear less frequently, and LC Z is rare. Overall, the dataset contains a large proportion of higher-quality fixes but also a substantial share of unquantified-error locations.

4 Visual Mapping for shark moving

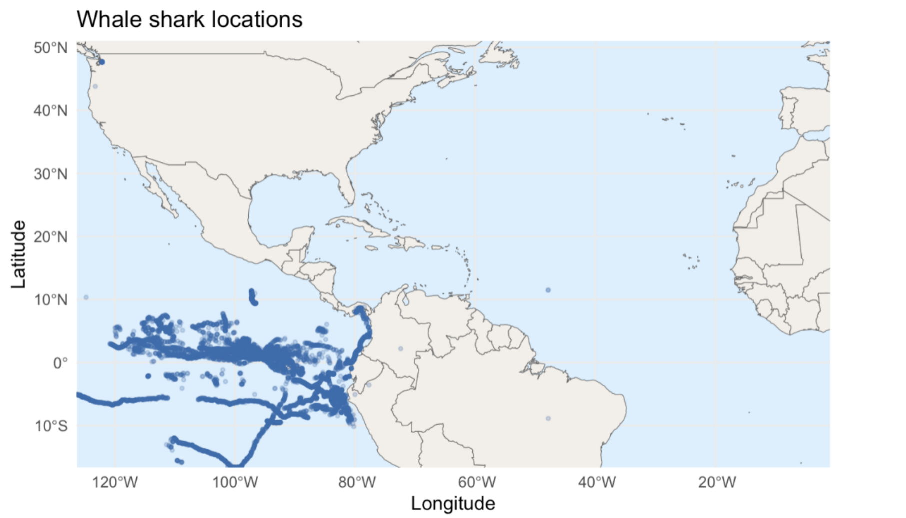

Figure 3: Tagged whale shark locations.

According to the shark location visualization, some locations appear to "cross continents", but these are most likely artifacts of Argos' location error rather than true movement. Argos position estimates can have heterogeneous measurement error depending on transmission conditions and the location class (lc). In this case, these abnormal observations are caused by the meausurement error.

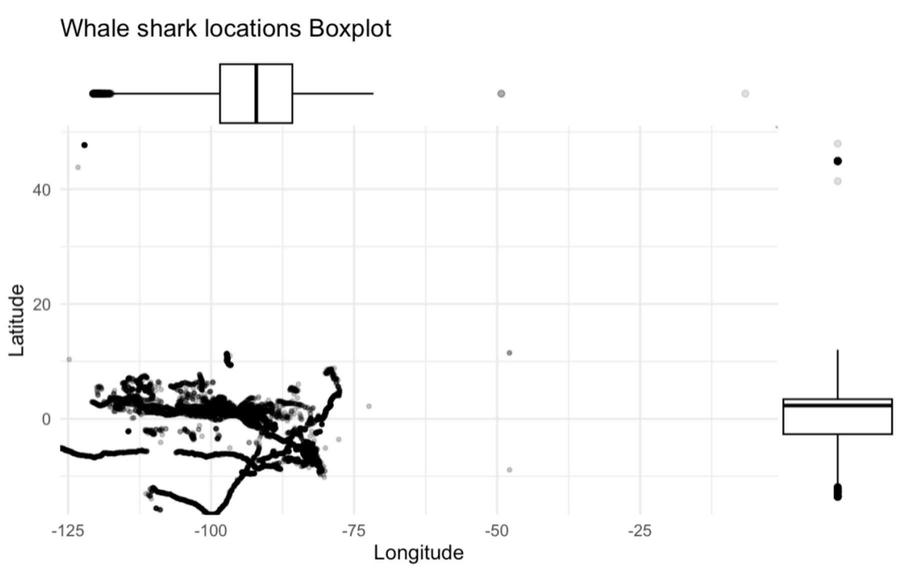

Figure 4: Whale shark locations boxplot

In this boxplot visualization, it shows that the tagged whale sharks' locations are highly clustered in a specific region. The main concentration is in the eastern Pacific around 120°W to 80°W and near equatorial latitudes around 15°S to 10°N. The mean latitude and longitude of shark movement is -0.628° and -96.03°. The dense point cloud and track-like streaks suggest repeated detections within this core area, indicating that most recorded movement occurs within a relatively constrained latitudinal band close to the equator. The marginal boxplots reinforce this pattern: longitude values are tightly centered on the eastern Pacific range and latitude values are centered near 0°, with a moderate spread to the south. A small number of extreme outliers appear far from the main cluster, which may reflect r potential location errors.

5 Analysis of the time gap among observations

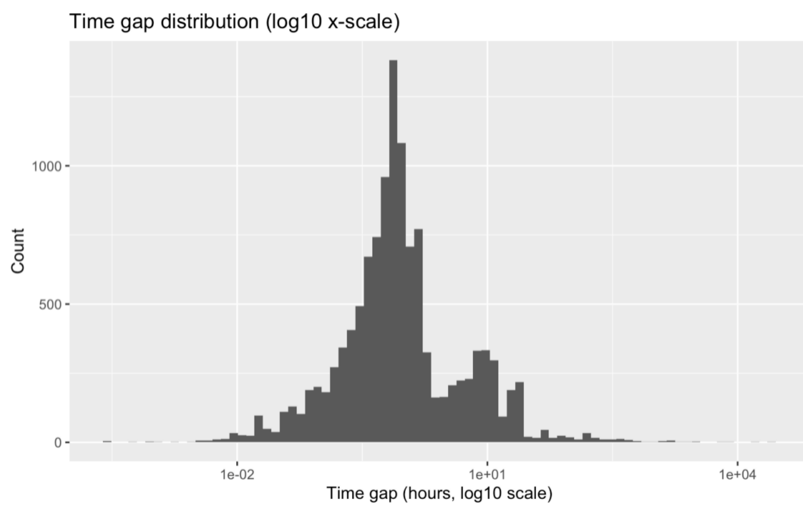

Figure 5: Time gap distribution between consecutive observations.

The telemetry observations are recorded at irregular sampling intervals; the time gap between consecutive locations is not constant across tracks. This leads to a strongly right-skewed, long-tailed distribution: most observations are separated by short intervals, but occasional long gaps occur.

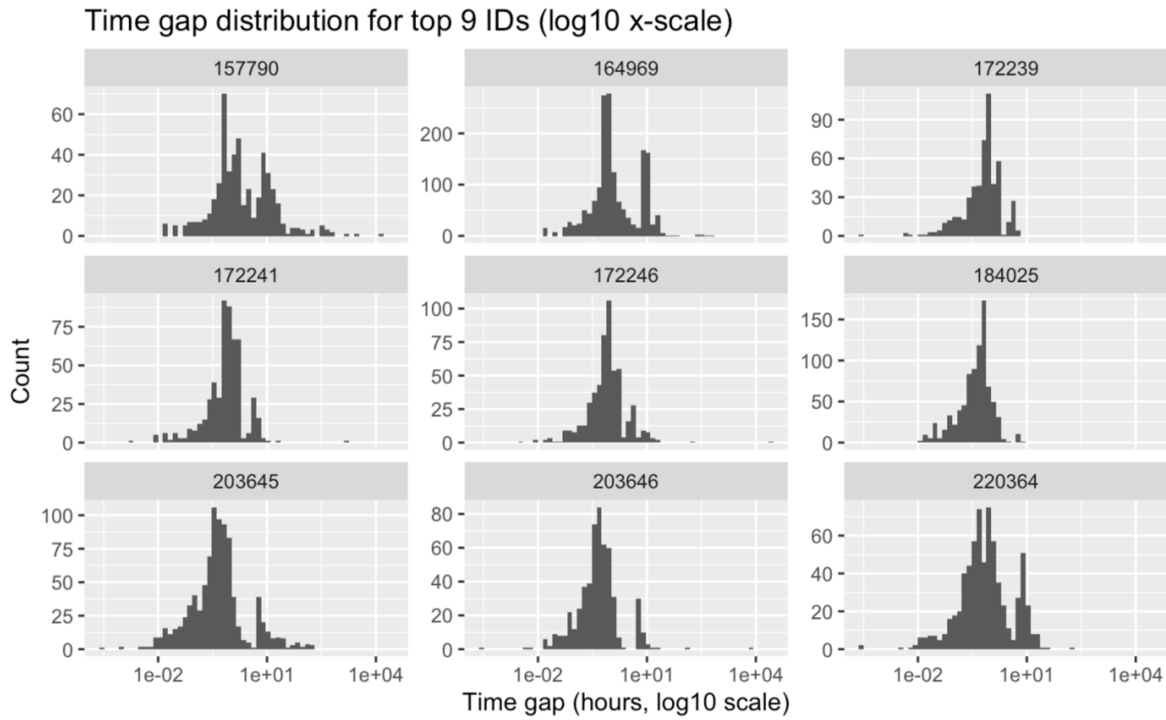

Figure 6: Time gap distributions for top 9 IDs.

The histograms show that time-gap distributions vary substantially across individuals, even among the top 9 IDs with the most observations. Although all tracks exhibit irregular sampling (with gaps spanning multiple orders of magnitude on the log10 scale), the shape and concentration of gaps differ by ID: some individuals have tightly clustered gaps, while others show broader, more right-skewed distributions with pronounced long gaps. This between-individual heterogeneity implies that sampling effort and data quality are not uniform across tags.

6 Outlier location(exceeding the maximum speed)

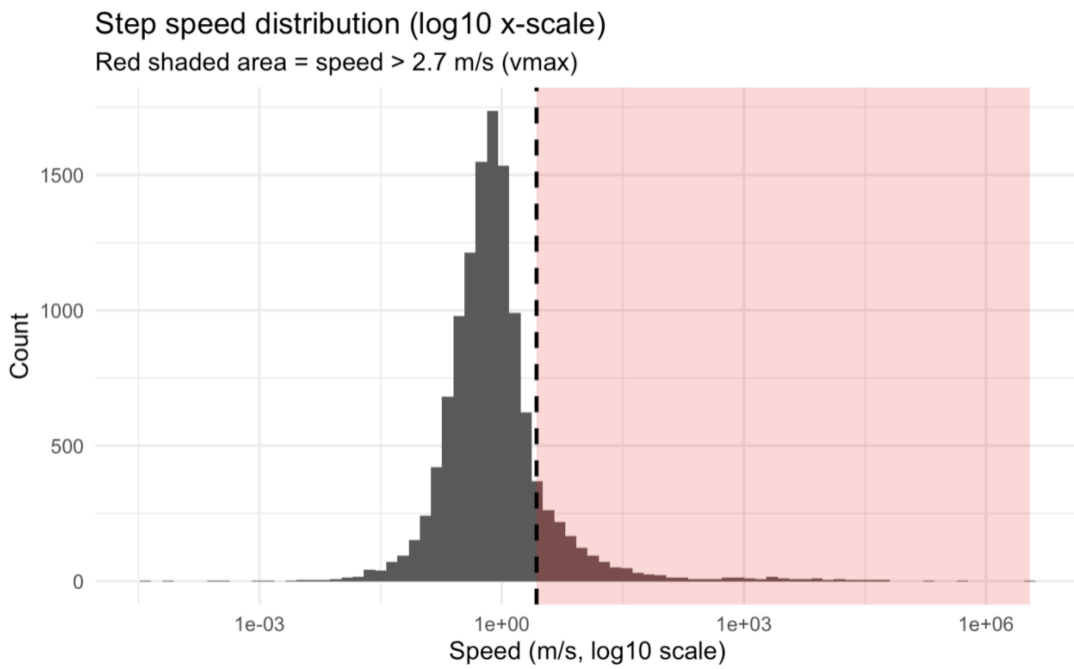

Figure 7: Step speed distribution (log10 x-scale) with vmax threshold.

This figure shows the distribution of step speeds (m/s) computed between consecutive locations within each individual. Because satellite telemetry can include Argos location errors, some consecutive fixes imply unrealistically fast movement. We use 2.7 m/s as a biological plausibility threshold for whale sharks; the dashed vertical line marks this cutoff, and the red shaded region highlights speeds exceeding it. Most steps fall well below 2.7 m/s, while a small right-tail extends beyond the threshold, indicating the outlier locations that would require biologically impossible travel rates.

7 Conclusion

To conclude, the whale shark telemetry dataset contains 12,173 locations from 60 tagged individuals from 2011 to 2021, but the data are highly uneven across sharks. The exploratory analyses show three key characteristics of the data. First, sampling is strongly irregular: time gaps are long-tailed and differ substantially across individuals. Second, location quality is heterogeneous: Argos location classes include many high-quality fixes but also a substantial fraction of A/B locations with unquantified accuracy, implying that observation error must be handled explicitly. Third, the spatial distribution is concentrated in the eastern Pacific near equatorial latitudes, while a small number of extreme outliers and a right tail of step speeds above the biological threshold of 2.7 m/s indicate occasional implausible locations that should be filtered. Taken together in the EDA, it motivates using a state-space modeling framework that accommodates uneven sampling, incorporates observation uncertainty (e.g., via LC), and applies a biologically informed maximum travel rate (vmax = 2.7 m/s) to identify and mitigate outlier locations before interpreting movement patterns.